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A Prospective Comparative Randomized Clinical Trial of Three Endodontic Retrograde Filling Materials

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Mineral trioxide aggregate (MTA) was first introduced by Mahmoud Torabinejad in 1993¹ for retrosealing roots following periapical curettage as a biocompatible replacement for the less effective materials that had been used throughout most of the 20th century, such as amalgam, IRM, Super EBA, etc.¹ Within the last 10 years, biocompatible calcium silicate materials (Biodentine and EndoSequence) have been introduced as replacements for MTA. The purpose of this prospective comparative randomized clinical trial was to determine which material was most effective in patients with periapical lesions of endodontic origin who underwent apical microsurgery and received one of three retrograde filling materials: EndoSequence (Brasseler, Savannah, Ga.), ProRoot MTA (Dentsply Sirona, Tulsa, Okla.) or Biodentine (Septodont, Saint-Maur-des-Fossés, France).

Materials and Methods

Brief Overview of the Study

This prospective comparative randomized clinical trial was conducted to evaluate the extent of healing following the use of three retrograde filling materials in patients who underwent apical microsurgery for the treatment of periapical lesions of endodontic origin. The participants received one of three retrograde filling materials: EndoSequence, ProRoot MTA or Biodentine. Patients returned for clinical and cone beam computed tomography (CBCT) assessments for up to six months after the procedure.

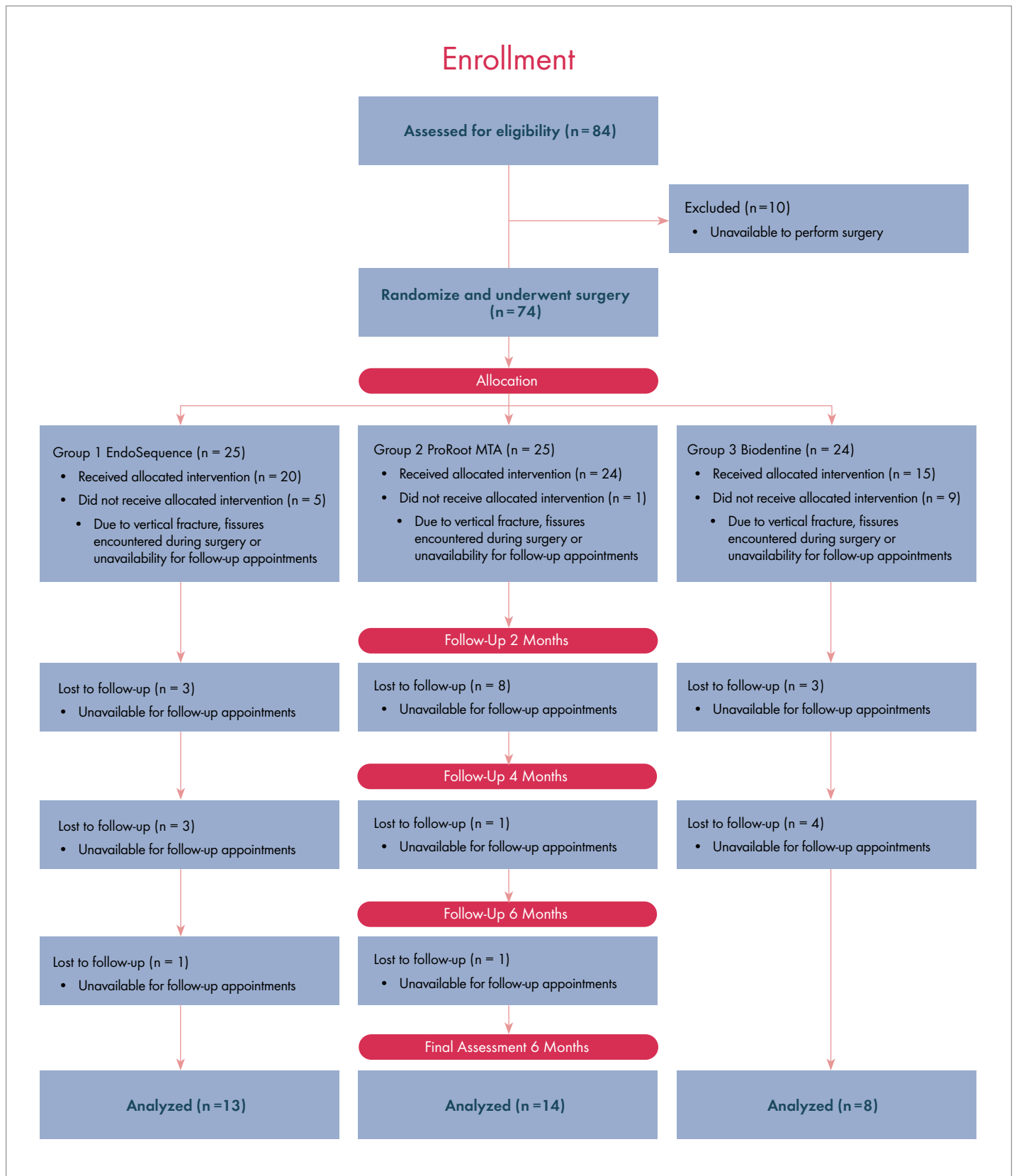
The patients were selected from the endodontics department at the University of Valparaíso, Chile. This endodontics department receives referrals from other

departments within the university, peripheral hospitals and health centers, private practices and the Chilean Navy Dental Center. The investigation was coordinated by Finis Terrae University in Santiago, Chile, which was responsible for approving the protocols and study design. The University of Valparaíso and the Chilean Navy in Valparaíso, Chile, were responsible for conducting the screenings and surgical procedures and the clinical and radiographic controls. The study was approved by the Institutional Review Board of Finis Terrae University, Resolution nr. 15/2017, Santiago, Chile.

A total of 74 patients underwent surgery. Fifteen patients were eliminated after surgery due to vertical root fracture or fissures encountered during the surgical procedure or because of unavailability to perform the follow-up assessments. Of the resulting 59 eligible patients, 51 patients completed two months of follow-up, 45 patients completed four months of follow-up and 35 patients completed six months of follow-up. Details about the sequence of patient selection and dropout are specified in **FIGURE 1**, following Consolidated Standards of Reporting Trials (CONSORT) guidelines.² The 59 patients were allocated for intervention based on block randomization in three groups (E = EndoSequence, M = ProRoot MTA, B = Biodentine) using an unpredictable random sequence by means of a software-based algorithm (www.random.org).

Patient inclusion criteria³ were adult patients, American Society of Anesthesiologists (ASA) classification I and II; any tooth with a single chronic periapical lesion requiring apical microsurgery; and informed consent signed prior to surgery.

Patient exclusion criteria were a background of drug or alcohol abuse, adjacent periapical lesion and periodontal disease.

**FIGURE 1.** Patient intervention flowchart.

Laboratory tests prior to surgery included a complete blood count, partial thromboplastin time, prothrombin time, fasting plasma glucose level and bleeding time.

Patient Screening

At the first screening appointment, all patients received a thorough clinical examination and further explanation of the surgical procedure from the endodontist. If eligible, the patient was asked to be part of this prospective comparative study; if the patient agreed, written instructions with the rules and conditions were given. Subsequently, the patient signed the informed consent and was incorporated into the study. Participants then completed a preoperative CBCT exam in the radiology department at the Chilean Navy Dental Center.

Surgical Procedure

The surgical procedure was performed by a single trained endodontist assisted by endodontic residents and assistants. Each procedure was performed in a state-of-the-art surgical room under a clinical microscope using ultrasonic handpieces and specialized microsurgical instruments.

Three biomaterials were used for this study: EndoSequence, ProRoot MTA and Biodentine.⁴⁻⁷

The material for each procedure was randomly selected by an assistant following the generated random sequence and mixed (if necessary) according to the manufacturer's instructions and out of the sight of the surgeon; the material was then handed to the surgeon for placement in the Class 1 apical preparation. Although randomly selected, the identities of some retrograde filling materials were evident upon presentation and packaging. Therefore, at this point, the participants were blinded from

TABLE 1

Surgical Protocol Applied to All Interventions

1. Premedication with amoxicillin 2 g and meloxicam 15 mg (or azithromycin 500 mg if patient is allergic to amoxicillin) one hour before the apicoectomy.^{8,9}
2. Local anesthesia with lidocaine HCl 2% 1:100,000 epinephrine.
3. Full-thickness flap in alveolar mucosa below the level of the tooth apex using a No. 15 scalpel blade.^{10,11}
4. Minimal osteotomy (no greater than 5 mm) using a high-speed handpiece and new carbide burs at 45 degrees under continuous normal saline solution irrigation.
5. Initial curettage using micro curettes.
6. 3 mm apicoectomy with 0- to 10-degree angulation using carbide-tapered burs (Zekrya, Dentsply Sirona, Charlotte, N.C.).
7. Fragment and remnant examination using methylene blue staining for root outline.
8. Additional curettage if necessary and hemostatic control.
9. 3 mm retrograde preparation with ultrasonic points (ProUltra Surgical, Dentsply Sirona).¹²⁻¹⁷
10. Drying of the surgical site using sterile paper points.
11. Retrograde filling using the randomly selected retrograde filling material.
12. Flap reposition and suture stitches using 4.0 silk.
13. Local compression and immediate ice application.
14. Scheduled follow-up appointment for suture removal in seven days.^{17,18}

material selection, but the surgeon was not blinded from the material, which was visibly different from others. The surgical protocol detail can be found in **TABLE 1**.

Data Acquisition and Follow-Up

After the microsurgery was completed, patient information was entered into a Microsoft Excel spreadsheet and sent to CBCT postoperative control at the radiology department of the Chilean Navy Dental Center. The date of each procedure and the patient data were sent to the study coordinator without revealing the material used; the identity of the material was kept blind until the results and statistical analysis were completed. To assure randomization of each patient without revealing names or personal data, a six-digit number was assigned to each patient using an online number generator (www.random.org). Therefore, both the radiologist and the biostatistician were blinded from the patient identity and the material used. The patients were called back after two,

four and six months for clinical and CBCT assessments. The DICOM archives were extracted every two weeks until final completion, and the data for the completed cases were sent via Dropbox to an oral radiologist at the University of Iowa in Iowa City, Iowa, for radiographic analysis and statistical comparison.

Five CBCT scans were acquired on a 3D Accuitomo 170 machine (Morita, Tokyo). The scans were acquired at a field of view of 4.2 cm by 4.2 cm at 0.13 mm/slice resolution with the voltage at 80 kVp and a tube current of 5 mA. CBCT scans were acquired preoperatively, postoperatively and at two, four and six months.

The CBCT scans were evaluated using Invivo6 software (Anatomage Inc., Santa Clara, Calif.); multiplanar reconstructions (MPR images) with cross-sectional images of the site of interest were evaluated. Scans were repositioned to ensure similar image orientation prior to measurements and evaluations of the areas of interest to prevent errors



FIGURE 2. CBCT scans; example of one procedure on tooth No. 6. Preoperative CBCT sagittal view showing the radiolucent area (**2A**). Postoperative CBCT sagittal view showing the radiolucent area with a defect in the bone area where the surgery was performed (**2B**). Two-month postoperative CBCT sagittal view showing early evidence of remineralization (**2C**). Four-month postoperative CBCT sagittal view showing significantly more remineralization (**2D**). Six-month postoperative CBCT sagittal view showing continuing remineralization; the outline of the original radiolucent area has been preserved showing the extent of the healing (**2E**).

due to variability in patient orientation during the process of image acquisition. All images were evaluated by a board-certified oral and maxillofacial radiologist with more than 15 years of experience.

The size of the periapical radiolucent area associated with the treated tooth was measured in each of the five scans.

In light of the relatively small sample size of the patient population and to prevent bias during the process of evaluation of measurements on the scans, the measurements of the radiolucent area were performed with at least a one-week interval between any two scans in the same patient. All evaluations were done twice to assess intraobserver reliability. The radiolucent area was measured to obtain the largest dimension of the lesion. Each dataset was scrolled through multiple times to accurately judge the largest possible dimension in any of the three directions (axial, coronal and sagittal views) and correlated with measurements on the cross-sectional views.

The radiologist was fully blinded as to which material was used for treatment in all of the patients.

Measurements were made on each scan; the extent of healing was evaluated subjectively on each of the postoperative scans and all data were tabulated. Subjective evaluation of healing was classified as follows (**FIGURES 2A–2E** for healing progress in an example case):

- Radiolucent with no evidence of healing (e.g., see initial preoperative or postoperative scans in **FIGURES 2A** and **2B**).

- Some degree of healing within the radiolucent area (e.g., see two-month interval scan in **FIGURE 2C**).
- Moderate degree of healing within the radiolucent area.
- Good healing with minimum evident radiolucent area (e.g., four-month and six-month scans for the same case shown in **FIGURES 2D** and **2E**).

In the initial postoperative cases, performed immediately following the procedure, all readings were classified as radiolucent with no evidence of healing, as expected (i.e., they showed low density).

Although subjective, the original outline of the lesion could be assessed in each of the postoperative scans even though the scans demonstrated evidence of progressive osseous healing at different time intervals.

groups. The Kruskal-Wallis test was used to examine differences among the groups and pairwise comparisons were conducted by using Mann-Whitney tests.

Conclusion

At the end of six months, there were no significant clinical differences between the three retrograde sealing materials used for this randomized double-blind investigation. If circumstances permit, we recommend using this study design with a 12-month follow-up. ■

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